# PMx Weekly Digest: November 03, 2026

**Content Source:** PMC Full Text

## Paper 1: Identifying Dose-Relevant Covariates With SCOUT: A Case Study of Lorlatinib.

### Bibliographic Information

- **Title**: Identifying Dose-Relevant Covariates With SCOUT: A Case Study of Lorlatinib.
- **Authors**: Ideno Yasuhisa
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology
- **DOI**: 10.1002/psp4.70355

**Paper Type**: POPK_PKPD

### Quick Take

This paper applies the SCOUT framework to a published lorlatinib PopPK model, shifting covariate analysis from PK parameters to the individualized optimal dose. It finds that sex, age, and renal impairment are associated with dose requirements, with sex explaining the largest variance. Pharmacometricians interested in precision dosing and covariate relevance should read this for a novel dose-oriented perspective.

---

### Executive Summary

The study reanalyzes a published lorlatinib PopPK model using the SCOUT framework to estimate individualized optimal daily doses for 330 NSCLC patients, targeting a trough concentration of 150 ng/mL at day 21. The estimated doses varied widely (mean 194 ± 134 mg/day), and exploratory analyses identified sex, age, and renal impairment as associated with dose requirements. LMG-based relative importance analysis showed sex explained 24.3% of the variance in optimal dose, with a bootstrap 95% CI of 9.84%–35.5%. The findings suggest that fixed dosing may not fully account for interindividual variability, supporting further investigation of precision dosing strategies for lorlatinib.

---

### Scientific Context & Motivation

Conventional covariate analyses in PopPK focus on statistical associations between patient characteristics and PK parameters (e.g., clearance), but do not directly address whether these effects translate into clinically meaningful dose differences. The SCOUT framework reframes the question by targeting the individualized optimal dose itself, bridging the gap between pharmacometric modeling and actionable dosing decisions. This study applies SCOUT to lorlatinib, a targeted therapy with known exposure-toxicity relationships, to explore which patient characteristics are associated with dose requirements, addressing the clinical need for precision dosing in oncology.

---

## ⚡ Methodological Snapshot

The study used a published PopPK model of lorlatinib (ADVAN13 implementation) to obtain empirical Bayes estimates of individual PK parameters. Individualized optimal doses were then estimated using the SCOUT framework, which minimizes an OFV that penalizes sub-target trough concentrations ($C_{target}$ = 150 ng/mL) over days 18-21 while also accounting for exposure. Associations between estimated doses and covariates were explored via visual inspection and LMG-based relative importance analysis with bootstrap confidence intervals.

---

## ⌧ Structural Model Breakdown

The structural model is a previously published PopPK model for lorlatinib, likely a two-compartment model with time-varying clearance. The original model used ADVAN4 (two-compartment) and was modified to ADVAN13 for this study. The exact compartments and parameters are not detailed in the abstract, but the model includes central and peripheral compartments, clearance (possibly time-varying), and intercompartmental clearance. The OFV for dose optimization integrates over days 18-21, with a penalty for trough concentrations below 150 ng/mL and a term proportional to exposure.

---

### Detailed Methodological Analysis

#### Modeling Approach

Population pharmacokinetic model with time-varying clearance (from reference [7]), implemented in NONMEM 7.5.1 with ADVAN13 for flexible differential equations. Individual parameters obtained as empirical Bayes estimates.

#### Data Sources

Anonymized pharmacokinetic data from the Supporting Information of the published PopPK study: 5806 plasma measurements from 425 individuals (patients and healthy participants). Only patients (PTST=1) were retained for dose estimation (n=330).

#### Estimation Methods

Empirical Bayes estimates from the fitted PopPK model; dose optimization via OFV minimization.

#### Model Evaluation

Consistency of parameter estimates with original implementation confirmed; sensitivity analyses for relative importance (Figures S1 and S2).

#### Covariate Analysis

Exploratory visual inspection and linear regression with LMG relative importance; bootstrap (n=1000) for confidence intervals; CYP phenotypes excluded due to missing data.

---

**Statistical Rigor Assessment**

The analysis is exploratory, not confirmatory. The LMG method provides a robust partitioning of variance independent of covariate ordering, with bootstrap confidence intervals. However, the reliance on post hoc estimates and a single target trough concentration limits statistical rigor. The sample size (n=330) is adequate for exploratory analyses, but missing data for genetic covariates reduced power. Sensitivity analyses were performed, but no formal hypothesis testing was conducted.

---

## ☒ Key Findings

Individualized optimal doses were successfully estimated for all 330 patients, with a wide distribution (mean ± SD: 194 ± 134 mg/day). Visual inspection suggested lower optimal doses in older patients, female patients, and those with greater renal impairment. Multivariate LMG analysis explained 25.5% of the variance in optimal dose, with sex being the most important covariate (relative importance 24.3%, 95% CI 9.84%–35.5%). These findings suggest that fixed dosing may not fully capture interindividual variability in exposure, potentially impacting tolerability given the known exposure-toxicity relationship.

---

## 💡 Clinical & Regulatory Implications

The findings suggest that fixed dosing of lorlatinib may not fully account for interindividual variability, potentially impacting tolerability given the exposure-toxicity relationship. Sex, age, and renal function may be relevant for dose individualization, but confirmatory studies are needed. The SCOUT framework provides a quantitative foundation for future precision dosing investigations, but no immediate dose-adjustment recommendations are made.

---

**Strengths & Limitations**

**Strengths**

- Novel application of SCOUT framework to a real-world PopPK model, shifting focus to dose relevance.
- Use of a published, well-established PopPK model with time-varying clearance.
- Transparent methodology with NONMEM control streams and data snippets provided.
- Robust sensitivity analyses (Figures S1 and S2) confirming the relative importance results.
- Clinically relevant target trough concentration based on exposure-response data.

**Limitations (Acknowledged by Authors)**

- Reliance on a previously published PopPK model and post hoc individual estimates, potentially affected by model structure and shrinkage.
- Uniform target $C_{trough}$ of 150 ng/mL applied across all patients, ignoring potential variation by disease stage or treatment context.
- Genetic covariates excluded due to substantial missing data, limiting assessment of metabolic enzyme polymorphisms.

- Lack of information on menopausal status and hormonal factors, preventing evaluation of sex-related differences.

**Limitations (Expert Review)**

- The OFV equation as extracted appears malformed; the penalty term likely should be $\rho^2 \cdot \max\left(0, C_{target} - C_{central}\right)^2$, but the provided text is ambiguous.
- The SCOUT approach relies on a single target trough concentration, which may not capture the full exposure-response relationship.
- The analysis is exploratory and does not establish causal relationships or actionable dose-adjustment rules.
- The LMG method assumes linearity and additivity, which may not hold for all covariate-dose relationships.
- The study does not account for potential time-varying covariates or adherence effects.

**Generalizability**

Findings are specific to lorlatinib and the studied NSCLC population; generalizability to other drugs or patient populations is limited. The SCOUT framework itself is generalizable, but results depend on the underlying PopPK model and target exposure.

---

**Key Equations**

**Objective Function Value (OFV) for Dose Optimization**

$$OFV = \int_{Day18}^{Day21} \left[\rho^2 \max\left(0, C_{target} - C_{central}\right)^2 + C_{central}\right] dt$$

Defines the objective function used to identify the minimal daily dose achieving the target trough concentration ($C_{target}$ = 150 ng/mL) over days 18-21, with a penalty for sub-target concentrations and a term proportional to exposure.

---

**Figures & Tables**

- **Figure 1**: Scatter plots and box-and-whisker plots showing relationships between individualized optimal dose and each covariate (continuous and categorical).
  - *Significance*: Visualizes univariate associations, suggesting lower optimal doses in older, female, and renally impaired patients.
- **Figure 2**: Relative importance estimates (LMG method) for each covariate in explaining variance in individualized optimal dose, with bootstrap confidence intervals.
  - *Significance*: Quantifies the multivariate contribution of each covariate, highlighting sex as the most important (24.3% of $R^2$).
- **Table S1**: Summary of patient characteristics for the 330 NSCLC patients included in the analysis.
  - *Significance*: Provides demographic and clinical context for the study population.
- **Figures S1 and S2**: Sensitivity analyses for the relative importance estimates.

- *Significance*: Demonstrates robustness of the LMG results across different analytical conditions.

---

**Code & Reproducibility Assessment**

The NONMEM control stream and a snippet of the data used for individualized dose estimation are provided in the Supporting Information. The R code for LMG analysis is not explicitly provided, but the relaimpo package is mentioned. The original PopPK model and data are from a published study (reference [7]).

---

**Supplementary Materials**

Supplementary materials include the NONMEM control stream, a snippet of the data used for dose estimation, a toy example illustrating the method, Table S1 (patient characteristics), and Figures S1 and S2 (sensitivity analyses).

---

**Future Directions**

Future studies should prospectively evaluate the identified covariates (sex, age, renal function) in dose-adjustment strategies. Integration with exposure-efficacy and exposure-safety data is needed to confirm clinical benefit. Genetic covariates (CYP phenotypes) should be included with complete data. The role of menopausal status and hormonal factors in sex-related differences warrants investigation. The SCOUT framework could be extended to other targeted therapies and to dynamic dosing strategies.

---

**Expert Commentary**

This paper represents a thoughtful shift in covariate analysis, addressing the clinically relevant question of dose requirements rather than PK parameters. The SCOUT framework is a welcome addition to the precision dosing toolkit. However, the reliance on post hoc estimates and a single target trough concentration limits the direct clinical applicability. The finding that sex explains a large portion of variance is intriguing but mechanistically unclear; future work should explore body size and CYP3A activity differences. The exploratory nature is appropriately emphasized, and the results should be viewed as hypothesis-generating. The provision of NONMEM control streams is commendable for reproducibility.

---

**Bottom Line**

For practicing pharmacometricians, this study demonstrates a practical application of the SCOUT framework to identify dose-relevant covariates, shifting the analytical focus from PK parameters to the individualized optimal dose. The results suggest that sex, age, and renal function may influence lorlatinib dose requirements, but the exploratory nature and reliance on post hoc estimates warrant caution. The approach offers a valuable template for future precision dosing investigations, but confirmatory studies and integration with exposure-response data are needed before clinical implementation.